Reducing Post-Harvest Losses of Tomatoes and Apples in Balochistan

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Executive Summary

Balochistan contributes significantly to Pakistan's agricultural output, particularly in the production of tomatoes and apples. Despite being a leading producer, over 40% of tomatoes and nearly all domestic apples. The region faces severe post-harvest losses due to climate change, inadequate cold storage, poor transport, and the absence of processing facilities. This report proposes the installation of localized processing units, funded via the Export Development Fund (EDF), to reduce wastage, stabilize prices, and foster rural employment.

Tomato Yield and Harvest

- **Annual Production**: 400,000–500,000 tonnes [5]
- **Harvest Seasons**: Spring and Summer
- **Major Producing Districts**: Lasbela, Barkhan, Chagai, Kachhi, Jhal Magsi, Khuzdar, Awaran, Kharan, Turbat, Panjgur, Gwadar, Washuk, Noshki

Tomato Wastage in Balochistan

- **Estimated Post-Harvest Losses**: 25–30% (120,000–150,000 tonnes annually) [3]
- **Core Reasons**:
    - Lack of cold storage
    - Absence of local processing facilities
    - Poor packaging and logistics infrastructure

Area-wise Tomato Yield

Table 1: District-wise Tomato Area and Production in Balochistan (2021–22 vs. 2022–23)
(Data sourced from the Ministry of National Food Security & Research [5])

District	Area 2021–22 (ha)	Area 2022–23 (ha)	Production 2021–22 (t)	Production 2022–23 (t)
Quetta	133	128	2141	512
Pishin	543	548	8704	6368
K. Abdullah	149	165	2384	2096
Chagai	89	90	1335	495

Noushki	117	119	1767	930
Loralai	537	590	8592	4320
Musa Khail	56	62	857	840
Barkhan	6187	3095	94250	45840
Zhob	52	54	842	528
Sherani	25	33	403	496
Killa Saifullah	2629	2629	38383	7800
Sibi	23	24	364	380
Harani	277	277	4404	4419
Ziarat	0	36	0	547
Kohlu	114	135	1752	2016
Dera Bughti	33	41	525	653
Nasirabad	2110	2110	37545	37569
Jaffarabad	525	620	9341	11049
Kachhi	380	435	5358	6145
Jhal Magsi	83	85	1266	1297
Kalat	874	878	15033	4233
Mastung	90	69	1521	170
Khuzdar	201	2505	3357	37176
Awaran	304	304	4429	4475
Kharan	78	232	1201	2865
Washuk	55	65	836	945
Lasbela	7740	7749	119977	120129
Turbat	127	135	1972	2101
Panjgur	136	167	2026	795
Gwadar	0	0	0	0
Total	**23667**	**23380**	**370565**	**307189**

District-wise Tomato Area and Production in Balochistan (2021–22 and 2022– 23)

Key Challenges in the Tomato Sector

- Inadequate cold storage and cold-chain systems
- Poor rural road infrastructure
- Short shelf life of fresh produce
- Lack of post-harvest packaging and processing [6]

Tomato Plant Machinery and Process Flow

Based on SMEDA's Pre-Feasibility Study [7]

1. Pre-Processing Equipment

- Sorting Belt Conveyor
- Washing Tub with Re-Circulation Pump
- Elevator and Bypass Conveyor

2. Core Processing Equipment

- Main Elevator

- Chopper, Pulper, Thermo Break
- Refiner for residual separation

3. Post-Processing & Storage

- Waste conveyor, tanks, and packaging in aseptic drums
- Storage in -20°C cold rooms

Expected Impact

- Reduction in tomato wastage
- Improved farmer income and price stability
- Increased employment and export opportunities

Tomato Plant Machinery Procurement

Both Local and imported plants can be selected for the project. Local fabricators use local and imported material and parts (available locally).But imported machinery is more reliable then local machinery in quality and results however local machinery or a mix of local and imported machinery can be selected for this plant.

Local machinery can be obtained from the following sources

For Vegetable and Fruit Processing

i. Installation & Fabrication Engineers (Private) Limited

21/22 K. M. Ferozepur Road, Lahore

ii. Unique Engineering works (Private) Limited,

233-S, Industrial Estate, Township, Lahore

iii. Central Engineering Services (Private) Limited,

119-S Industrial Estate, Township, Lahore

iv. Two Star

Industrial Estate, Township, Lahore

For Refrigeration Machinery

i. Pakistan Air-conditioning Engineering Co. (Private) Limited

Gardee Trust Building, Napier Road, Lahore

ii. Kold Kraft,

Industrial Estate, 247-S Kot Lakhpat, Lahore

Imported Machinery Suppliers

Imported Plant of Italian and Chinese origin is available comprising of main sections of the plant only, which is to be supplemented with certain local components. Offices of suppliers of foreign plants for paste and fruits pulp production machinery are located at:

i. Burtuzi, Lahore Cantt

ii. Tetra Pak, Upper Mall, Lahore

Local Machinery along with foreign components is recommended for good quality, better efficiency and lower cost.

Utilities Required

- Electricity
- Gas or Furnace Oil or Coal
- Water
- Telephone
- Fax

Tomato Plant Machinery cost breakdown

US$200,000 = 57,000,000pkr [8]

Area Required for Project (Project Plan)

Required Land 108,000 (3 Acre) [7]

Suggested Location for the Processing Plant

After analyzing district-wise tomato production data and evaluating post-harvest loss factors, Lasbela district emerges as the most viable and strategic location for establishing a tomato processing unit in Balochistan. Lasbela recorded the highest tomato output (120,129 tons in 2022–23), contributing over 39% of the province's total production.

Key advantages of selecting Lasbela:

- Proximity to Karachi, Pakistan's largest metropolitan market and export hub, enabling reduced transportation time and cost.
- Favorable climate with fewer extreme weather disruptions, ensuring year-round operational feasibility.

- Established transport infrastructure, including the RCD Highway and ongoing CPEC developments, which ensure smoother supply chain movement.
- Lower risk of wastage due to its geographical closeness to the production clusters in Awaran, Khuzdar, and Kharan.
- Availability of semi-skilled labor and opportunity for rural employment creation.

Should Lasbela be logistically or financially unfeasible, the second-best option would be Khuzdar, which witnessed a significant production jump from 3,357 tons in 2021–22 to 37,176 tons in 2022–23, suggesting a growing tomato belt and potential for scalability.

Working Time Estimates

Description Time	
Working days per month: Tomato Paste Manufacturing Fruits Pulp/Paste Manufacturing	24 16 hours in 2 shifts11 for 30 days 16 hours in 2 shift for 110 days
Working hours per month: Tomato Paste Manufacturing Fruits Pulp/Paste Manufacturing	 30 days (384 hours) 30 days (192 hours)

STRATEGIC FACTORS

Key Success Factors

- Effective quality control in operations
- Competitive price of end products
- Abundant supply of raw material
- Cost efficiency through better management
- Media campaign for the awareness of the retail customers
- Availability of low cost skilled labor

Threats

- Crop failure
- Influence of major local and foreign brands operating in the market
- Change in the Government regulations
- Shrinkage of retail customer market over a period of time
- Fruit and vegetable growers are usually unable to get right prices for their good quality fruits/vegetables. The year of good harvest gives them more loss because of low prices in the local market

Apple Production in Balochistan

- **Annual Production**: ≈480,000 tonnes on 35,000 ha [5]
- **Apple wastage**: Balochistan exceeds 100,000 tonnes annually.
- **Harvest Season**: August to October
- **Key Producing Districts**: Ziarat, Kalat, Pishin, Killa Saifullah [3]
- **Post-Harvest Losses**: 20–40% [4, 5]
- **Cold Storage Damage**: 17% crate damage; 12% unsellable [5]

Area-wise Apple Yield

Table 2: District-wise Apple Area and Production in Balochistan (2021–22 vs. 2022–23)
(Data sourced from the Ministry of National Food Security & Research [1])

District	Area 2021–22 (ha)	Area 2022–23 (ha)	Production 2021–22 (t)	Production 2022–23 (t)
Quetta	1341	1342	32496	11554
Pishin	6739	6754	61060	71967
K. Abdullah	5619	5552	67650	70025
Chagai	0	0	0	0
Loralai	1385	1157	9685	11246
Musa Khail	68	37	229	0
Barkhan	1169	1363	17038	18673
Zhob	6473	6451	68670	35295
Killa Saifullah	39235	38621	269843	462067
Sibi	0	0	0	0
Ziarat	3969	3993	53911	23495
Kohlu	56	64	1035	771
Dera Bughti	0	0	0	0
Nasirabad	0	0	0	0
Jaffarabad	0	0	0	0
Bolan	0	0	0	0
Jhal Magsi	0	0	0	0
Kalat	1728	1728	13397	23492
Mastung	2740	2739	74607	6139
Khuzdar	856	866	10827	6607
Awaran	328	331	4401	1860
Kharan	0	0	0	0
Lasbela	0	0	0	0
Turbat	0	0	0	0
Panjgoor	76	79	28	23
Gwadar	0	0	0	0
Washuk	0	13	0	0
Kachhi	0	0	0	0
Harnai	79	79	954	684

Sherani	56	49	445	228
Naushki	0	0	0	0
Total	**71917**	**71218**	**686276**	**744126**

District wise Area and Production in Balochistan

Challenges in Apple Sector

- Limited cold storage capacity (700 tonnes for entire province)
- Transport-related losses up to 28% [2]
- Poor market access due to road conditions
- No local grading or processing units
- Insufficient irrigation and climate-related stress

Proposed Solutions

- Establish processing plants for apple-based products (jams, juices, cider, vinegar)
- Prepare land and install pulpers, boilers, and allied machinery
- Target installation in high-yield districts like Pishin and Killa Abdullah

Conclusion: Feasibility and Benefits

Investing in tomato and apple processing units in Balochistan is both feasible and beneficial. By addressing post-harvest losses and value addition, the initiative would uplift rural economies, generate employment, and reduce food waste. With EDF support and collaboration with local manufacturers, the proposed strategy aligns with national food security and economic development goals.

Advantages of Installing a Processing Plant

1. **Reduces Post-Harvest Losses**
    - Up to 30% of tomatoes and 40% of apples go to waste annually due to inadequate preservation methods.
    - Processing units can convert excess or perishable produce into value-added products such as paste, pulp, juice, and jams.
2. **Value Addition Increases Farmer Income**
    - Processed products fetch 2–4x higher prices than raw produce.
    - Increases profitability for both small and large-scale farmers while encouraging quality cultivation.
3. **Price Stabilization**
    - Absorbs surplus produce during peak seasons, helping prevent market crashes.

    - Enables consistent income for growers and avoids price exploitation during off-seasons.

4. **Employment Generation**
    - Creates new jobs in processing, packaging, cold storage, transport, and equipment maintenance.
    - Encourages rural employment, particularly for youth and women.
5. **Boosts Exports and Regional Trade**
    - Processed goods are more viable for export due to longer shelf life and international-standard packaging.
    - Opens doors to regional markets in the Middle East and Central Asia.
6. **Government Support via EDF**
    - The **Export Development Fund (EDF)** can fully or partially finance plant installation, machinery procurement, and related infrastructure.
    - Aligns with national goals on food security, export diversification, and rural development.

Hurdles to Consider

1. **High Capital and Operating Costs**
    - Requires upfront investment in land, construction, processing equipment, and energy backup.
    - Ongoing costs involve trained manpower, electricity, packaging, and plant maintenance.
2. **Seasonal Crop Dependency**
    - Tomato and apple harvests are limited to specific months, leading to potential underutilization of the plant.
    - Diversifying the plant to process other fruits or vegetables is essential for year-round operation.
3. **Infrastructure Limitations**
    - Underdeveloped rural roads, unreliable electricity, and lack of cold storage hinder efficient supply and distribution.
    - Upgrades in logistics and local infrastructure are necessary to support plant sustainability.
4. **Market and Branding Challenges**

    - Without strong branding, processed products may struggle against established market players.
    - A robust marketing strategy is needed for domestic and international market penetration.
5. **Shortage of Skilled Workforce**
    - Local communities currently lack trained plant operators, food technicians, and packaging staff.
    - Training programs must be established alongside plant development.

Suggested Location for the Processing Plant

For apple production, the clear frontrunner is Killa Saifullah, producing a staggering 462,067 tons in 2022–23, accounting for over 62% of Balochistan's total apple output. Its centrality among key apple-growing districts (Pishin, Zhob, and Killa Abdullah) makes it an ideal processing and aggregation point.

Why Killa Saifullah is ideal:

- Largest and most stable apple production zone in the province, supported by a consistent increase in yield and cultivated area.
- Cold, high-altitude climate favorable for apple preservation and grading.
- Natural clustering with other major apple districts ensures consistent raw supply and minimizes input transportation costs.
- Opportunities for value-added processing such as juice, jam, cider, and vinegar, which can absorb seasonal surplus and reduce cold storage dependency.
- Creation of local employment and boosting of rural economic activity in a traditionally agricultural region.

While districts like Ziarat, Kalat, and Pishin are also key producers, their fluctuating yields and relatively weaker infrastructure make them more suitable as feeder districts to the central processing hub in Killa Saifullah.

Area Required for Project (Project Plan)

Required Land 108,000 (3 Acre) [7]

Average cost for a processing plant

US$200,000 = 57,000,000pkr [9]

Labour Requirement

Number of Officers and Staff required for processing the plant, the following administrative and production staff will be required

- Production Manager
- Quality control In charge
- Deputy Production Manager
- Shift In charge
- Production Supervisor
- Process In charge
- Boiler Engineer
- Electrical Engineer
- Mechanical Supervisor
- Chief Executive
- Manager Finance & Admin.
- Accounts Officer
- Administration Officer
- Accountant
- Accounts Assistant
- Computer Operator
- Peon
- Gardener
- Security Guards
- Marketing Staff
- skilled/semi skilled workers

Finalized Optimal Suggested Location

Quetta is recommended as the central location for establishing a combined tomato and apple processing plant, based on a comprehensive analysis of production clusters, road connectivity, and infrastructure readiness.

While Quetta's own production is moderate, it serves as a logistical hub connecting major high-yield districts such as Lasbela (tomatoes) and Killa Saifullah, Pishin, and K. Abdullah (apples). Its central location, relatively developed infrastructure, and availability of utilities and labor make it the most feasible choice for a shared processing facility.

This centralized approach helps reduce post-harvest losses, enables cost-sharing across crop types, and ensures year-round utilization by operating in seasonal processing cycles. It also facilitates smoother transport, better storage, and efficient export logistics through Quetta's proximity to national highways and urban markets.

Conclusion

Establishing tomato and apple processing plants in Balochistan is a feasible, timely, and impactful initiative that addresses persistent post-harvest losses and creates a foundation for economic growth. Despite logistical and operational challenges, the advantages clearly outweigh the hurdles. With support, targeted location planning, diversified crop integration, and local capacity building, this initiative can:

- Reduce national food waste
- Strengthen farmer profitability
- Create rural jobs
- Support Pakistan's food exports

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